Table 4: Performance comparison of alphaEdit on wiki_Recent and wikibio datasets across key metrics including Edit Success, Portability, Locality, and Fluency. Results are averaged with standard deviations. The best results are highlighted in bold.

Method	wiki_recent				wikibio		
	Edit Succ.↑	Portability↑	Locality↑	Fluency↑	Edit Succ.↑	Locality↑	Fluency↑
MEMIT	56.25$_{\pm 0.28}$	42.73$_{\pm 0.27}$	41.02$_{\pm 0.20}$	513.35$_{\pm 3.47}$	63.73$_{\pm 0.40}$	64.27$_{\pm 0.41}$	582.38$_{\pm 3.34}$
RECT	82.47$_{\pm 0.53}$	51.28$_{\pm 0.25}$	48.84$_{\pm 0.24}$	568.62$_{\pm 3.71}$	91.48$_{\pm 0.48}$	72.83$_{\pm 0.44}$	612.04$_{\pm 4.29}$
AlphaEdit	**96.10**$_{\pm 0.47}$	**57.30**$_{\pm 0.38}$	**54.76**$_{\pm 0.30}$	**594.52**$_{\pm 3.91}$	**95.34**$_{\pm 0.46}$	**75.34**$_{\pm 0.50}$	**618.35**$_{\pm 4.22}$

Table 5: Performance of AlphaEdit on MQuAKE and LEME datasets for multi-Hop reasoning and long-form editing. Metrics include Multi-hop Reasoning, Chain-of-Thought (CoT) Multi-hop Reasoning, Edit Consistency, Factual Consistency, and Internal Consistency. The best results are highlighted in bold.

Model	Method	MQuAKE		LEME		
		Multi-hop↑	Multi-hop(CoT)↑	Edit↑	Factual↑	Internal↑
GPT-J	MEMIT	3.35$_{\pm 0.07}$	6.13$_{\pm 0.12}$	2.11$_{\pm 0.18}$	2.02$_{\pm 0.17}$	3.84$_{\pm 0.29}$
	RECT	3.77$_{\pm 0.04}$	7.61$_{\pm 0.20}$	2.24$_{\pm 0.20}$	2.62$_{\pm 0.19}$	4.07$_{\pm 0.31}$
	AlphaEdit	**5.03**$_{\pm 0.16}$	**9.14**$_{\pm 0.21}$	**3.34**$_{\pm 0.26}$	**3.80**$_{\pm 0.28}$	**5.42**$_{\pm 0.41}$
GPT2-XL	MEMIT	3.14$_{\pm 0.08}$	6.25$_{\pm 0.11}$	1.92$_{\pm 0.22}$	2.31$_{\pm 0.20}$	3.85$_{\pm 0.34}$
	RECT	3.72$_{\pm 0.06}$	7.48$_{\pm 0.24}$	2.12$_{\pm 0.26}$	2.60$_{\pm 0.21}$	4.13$_{\pm 0.29}$
	AlphaEdit	**5.00**$_{\pm 0.23}$	**9.25**$_{\pm 0.27}$	**3.28**$_{\pm 0.36}$	**3.07**$_{\pm 0.33}$	**5.76**$_{\pm 0.49}$

96.10% editing success, which is significantly higher than the second-best method, RECT (82.47%). A similar trend is observed on wikibio, where AlphaEdit reaches 95.34%, outperforming RECT by a substantial margin.

- AlphaEdit demonstrates the best performance in both Multi-hop and Multi-hop (CoT) tasks, with scores of 9.14 and 9.75 respectively, significantly surpassing competing methods. This showcases its strong capability in handling complex reasoning tasks and ensuring logical consistency across interdependent facts.
- On LEME, AlphaEdit excels in all three metrics, showcasing its ability to generate accurate long-form outputs. Its consistently high performance across GPT-J and GPT2-XL reinforces its reliability in executing precise edits while maintaining the structural coherence and integrity of the generated text.

C.8 Impact of Dataset Size on AlphaEdit's Performance

To explore the relationship between dataset size for calculating $\boldsymbol{K}_0$ and AlphaEdit's performance, we conduct additional experiments to evaluate the robustness of the model under reduced dataset conditions. Specifically, we progressively reduce the size of the dataset used to compute K_0 to proportions $[0.9, 0.8, 0.7, \ldots, 0.1]$ of its original size. The goal is to analyze the impact of dataset size on three key metrics: Efficacy, Generalization, and Specificity. All the results are summarized in Table 6. To further illustrate trend changes, we select the results for LLaMA3 and visualized them using line charts, as presented in Figure 11. According to Figure 11 we can find that:

- As the dataset size decreased, both Efficacy and Generalization demonstrate notable stability. Even at only 10% of the original dataset size, the drop in these metrics is negligible (less than 5%), suggesting that AlphaEdit effectively generalizes to unseen data and remains efficient even with reduced data availability.
- In contrast, the Specificity metric experience a significant decline as the dataset size is reduced. When the dataset size is limited to just 10% of its original volume, Specificity drop by 11.76%, indicating that the model's ability to store neighborhood knowledge heavily relies on the availability of a sufficiently large dataset.

Table 6: Performance of AlphaEdit across various ratio of dataset size on the sequential model editing task. *Eff.*, *Gen.* and *Spe.* denote Efficacy, Generalization and Specificity, respectively.

Model	Ratio	Counterfact			ZsRE		
		Eff.↑	**Gen.↑**	**Spe.↑**	**Eff.↑**	**Gen.↑**	**Spe.↑**
LLaMA3	1.0	98.90±1.21	94.22±0.89	67.88±1.34	94.47±0.97	91.13±1.02	32.55±1.78
	0.9	98.32±0.92	93.87±1.56	66.23±1.18	94.12±1.44	91.76±1.02	31.89±1.23
	0.8	96.75±1.35	92.45±0.78	66.45±0.99	94.12±1.23	90.95±1.42	30.67±1.09
	0.7	95.66±0.76	93.11±0.98	64.89±1.41	93.87±1.11	90.45±0.97	29.34±0.84
	0.6	96.12±0.86	91.34±1.23	63.34±0.94	93.87±1.36	91.12±1.11	29.34±1.25
	0.5	97.93±1.23	94.01±1.09	63.51±0.97	92.96±0.89	91.67±1.03	28.56±1.34
	0.4	95.88±0.78	92.67±1.11	61.78±1.09	92.98±1.09	91.76±1.28	28.56±0.99
	0.3	96.98±1.67	93.22±0.99	58.56±1.08	93.12±1.43	89.12±1.23	27.56±1.67
	0.2	97.45±0.97	91.89±1.22	56.89±0.89	93.01±0.84	89.99±1.09	26.34±1.34
	0.1	95.21±1.03	90.12±1.45	56.12±1.22	92.01±1.02	89.97±1.28	25.89±1.47
GPT2-XL	1.0	99.50±0.98	93.95±1.13	66.39±0.89	94.81±1.56	86.11±1.24	25.88±1.42
	0.9	97.82±1.43	92.78±0.87	65.24±0.92	93.67±1.05	85.73±1.12	25.05±1.32
	0.8	98.47±1.12	92.54±1.32	64.89±0.76	93.21±0.99	85.48±0.78	23.98±1.24
	0.7	96.23±1.09	93.21±1.24	64.45±0.98	94.05±1.09	85.74±0.89	23.67±1.11
	0.6	99.12±0.87	91.33±1.07	64.45±0.93	94.31±0.99	84.85±1.45	23.45±0.98
	0.5	95.68±0.92	90.89±1.34	61.76±0.76	94.74±0.87	85.92±1.23	22.78±0.76
	0.4	97.54±1.45	91.76±1.23	60.23±1.09	93.52±0.98	84.45±0.88	22.34±1.01
	0.3	99.01±1.09	90.32±1.11	58.92±0.92	93.01±1.34	83.78±0.99	22.67±1.21
	0.2	95.89±0.78	91.03±1.03	58.14±1.23	93.04±1.09	85.07±0.95	22.45±1.15
	0.1	96.35±1.02	91.03±1.32	58.14±1.14	93.04±1.22	85.07±1.43	21.11±1.12
GPT-J	1.0	99.75±1.15	96.38±1.45	75.48±1.25	99.79±1.28	96.00±1.67	28.29±1.32
	0.9	99.43±1.09	96.14±1.08	74.75±1.11	97.63±1.03	96.11±0.98	27.12±1.43
	0.8	98.34±0.76	96.03±1.12	75.21±0.89	97.65±0.87	96.01±1.32	25.89±0.89
	0.7	98.21±1.23	95.11±1.56	73.58±1.01	98.78±1.15	95.34±0.94	25.09±1.09
	0.6	98.94±1.17	95.33±1.12	73.89±1.21	99.05±1.09	95.45±0.84	24.87±1.45
	0.5	98.46±1.21	94.89±1.09	70.88±1.05	98.94±1.04	95.12±1.28	23.78±0.97
	0.4	97.74±0.88	94.76±1.04	69.89±1.18	98.31±1.23	94.91±1.09	23.67±1.22
	0.3	96.74±1.32	94.34±0.95	67.95±0.84	97.58±0.98	94.23±1.15	22.78±1.12
	0.2	96.94±1.09	94.73±1.24	68.04±0.92	97.34±0.97	94.12±1.02	23.45±1.09
	0.1	97.02±0.89	94.73±0.98	68.04±1.09	97.58±1.21	94.23±0.89	23.67±1.32

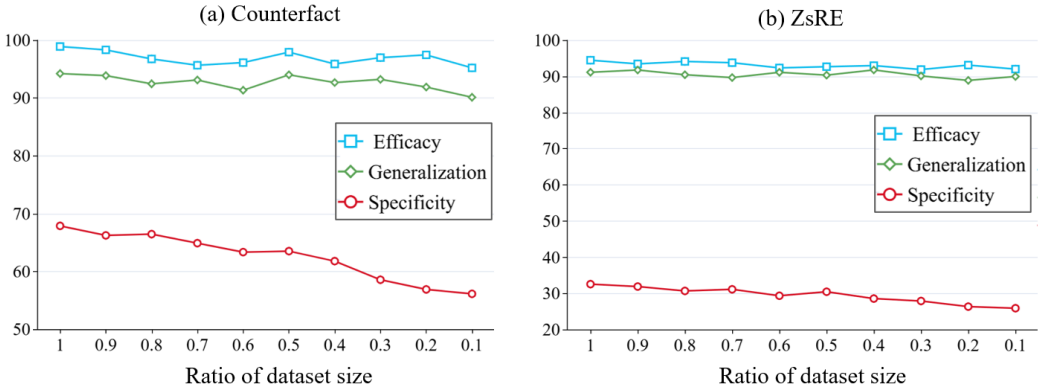

Figure 11: Performance of AlphaEdit across various ratio of dataset size on the sequential model editing task. Best viewed in color.

C.9 Runtime Evaluation of AlphaEdit

The computational complexity of the null-space projection in AlphaEdit depends solely on the dimension of the hidden dimension d_0 within the base LLM. Specifically, calculating the null-space

Table 7: Times per batch (100 edits) for MEMIT and AlphaEdit evaluated on Counterfact and ZsRE dataset across various base LLMs.

Method	**Counterfact**			**ZsRE**		
	LLaMA3	**GPT-J**	**GPT2-XL**	**LLaMA3**	**GPT-J**	**GPT2-XL**
MEMIT	222.51s	334.74s	474.14s	231.32s	344.21s	488.37s
AlphaEdit	223.24s	334.93s	476.79s	231.40s	345.52s	490.25s

projection matrix only requires operations on $\boldsymbol{K}_0\boldsymbol{K}_0^T \in \mathbb{R}^{d_0 \times d_0}$, which are independent of the number of layers, model size, or the knowledge base size.

To empirically validate the scalability of our method, we measured the average runtime for performing 100 edits with AlphaEdit and MEMIT on three LLMs with different model sizes and knowledge bases: LLaMA3, GPT-J, and GPT2-XL. The results are summarized in Table 7.

From these results, it is evident that AlphaEdit does not incur additional runtime overhead compared to MEMIT, even as the model size or knowledge base grows. This supports our claim that the null-space projection method is highly scalable and practical for large-scale model editing tasks.

D Visualizing the Counterfact and ZSRE Datasets Through Examples

To help readers unfamiliar with model editing tasks better understand the Counterfact and ZSRE datasets, we provide two examples from them in Figure 12 and 13. These examples illustrate the types of modifications and factual updates applied to the models during the editing process.